Name: _____
Surname: _____
Date: _____

Instructions

Circle the correct answer for each question, as each question has **only one** correct answer. Provide reasoning for your answer in the space next to each question.
A correct answer always awards a point (even with no explanation).
An incorrect answer with sensible reasoning awards half a point.
An incorrect answer with faulty or no reasoning subtracts one point.
No answer gives zero points (neither adds nor subtracts points).
Assume each code cell is **run in isolation**, meaning no other code has been run before it.

Questions

1. After running this code, what will be the final value of the **variable "x"**? And its type?

```
y = 3
x = 8
x += 1
x = x / y
```

   (a) 3 (type `int`)
   (b) 3.0 (type `float`)
   (c) 2.667 (type `float`)
   (d) 8/3 (type `real`)

2. What will be printed after running the following code?

```
message = "good soup"
message.upper()
message_new = message + "!!!"
print(message_new)
```

   (a) "good soup"
   (b) "GOOD SOUP"
   (c) "good soup!!!"
   (d) "GOOD SOUP!!!"

3. Which of the following lines of code will **raise an error**?

```
number = 5
add = int(number) + 2    # Line 2
text = str(number) * 10    # Line 3
print(f"{number} + 2 = 7")    # Line 4
```

(a) Line 2.

(b) Line 3.

(c) Line 4.

(d) None of them.

4. What needs to be fixed in the following code to **avoid an error**?

```
message = "This will always be a string"
if len(message) > 500.0:
  print("Too much text")
elif:
  print("Good!")
```

(a) Change `len(message)` to `message`.

(b) Change `500.0` to `500`.

(c) Change `elif` to `else`.

(d) The code works correctly as it is.

5. What will be **the last number** printed after running this code?

```
ls_fibonacci = [0, 1, 1, 2, 3, 5, 8, 13]

for num in ls_fibonacci:
  if num == 5:
    break
  print(num)
```

(a) 3

(b) 5

(c) 13

(d) The code will never stop running.

6. After running this code, what will be the final value of the **variable "result"**?

```
result = 10

for num in range(1, 4, 2):
  result = result - num
```

(a) 6

(b) 5

(c) 0

(d) The code will raise an error.

7. After running this code, what will be the final value of the **variable "u"**?

```
u = 6
v = 2

while u >= v:
  u = u - 1
  v = v + 1
```

(a) 6

(b) 4

(c) 3

(d) The code will never stop running.

8. After running this code, what will be printed?

```
def exercise(x: float) -> float:
    y = x * 2
    return y

exercise(8)
print(y)
```

(a) 8

(b) 16

(c) 16.0

(d) The code will raise an error.

9. Which one of the following statements about functions is **FALSE**?

   (a) Functions can be defined inside other functions.

   (b) There can be multiple return statements in a function.

   (c) Functions must always contain a return statement.

   (d) Type hints are not mandatory.

10. After running this code, what will be printed?

```
def exercise(singer="Taylor Swift", song="Blank Space"):
    msg = "My favorite song by " + singer + " is " + song
    print(msg)
```

   (a) Nothing will be printed.

   (b) Python will print "My favorite song by Taylor Swift is Blank Space".

   (c) Python will print "My favorite song by is ".

   (d) The code will raise an error.

Answers: 1-B, 2-C, 3-D, 4-C, 5-A, 6-A, 7-C, 8-D, 9-C, 10-A